Ordinary charged fields at fixed half filling

Fourier conditioning, internal contractions and the same-copy strong bridge

Alan Day Chengheng*

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Abstract

We extend the central-sector marked-source calculus to ordinary charged blocks with internal Gaussian contractions. Exact charge projection, with an integrable Fourier bound after singleton cancellation, subtracts $2\lambda(\sum_a i_a z_a)^2$ from the full-trace source exponent. Here i_a counts internal chords in charged block a. This removes the shared charge fluctuation and establishes the ordinary-to-Wick strong bridge in every fixed finite central Schatten norm. Exact energy pinching and fixed polynomial preparation bounds then transfer each fixed ordinary coefficient to the single-selected state. Uniform tails of the actual infinite ordinary series remain to be established.

1 Definitions and the missing conditioning step

Use standard CAR, a Hermitian overlapping complex Gaussian p-body Hamiltonian H with the same disorder in all fields, even N, m=N/2 and $\lambda_N = p^2/N \to \lambda > 0$. Let $C_Q = \mathbb{E}H^2$, with sector value C_m, and $h_N = H_m/\sqrt{C_m}$. Use normalized annealed sector norms $\|X\|_{r,m} = (\mathbb{E}\,\mathrm{tr}_m\,|X|^r)^{1/r}$. All degrees and source counts are fixed before N tends to infinity. Time uses dimensionless h units, with $\hbar = 1$.

The body loss and half-size maps are

$$DX = QX - \sum_i c_i^\dagger X c_i, \qquad \mathcal{L}X = \tfrac{1}{2}(NX - \sum_i (c_i^\dagger X c_i + c_i X c_i^\dagger)).$$

They act before sector restriction. Disjoint charged D marks may now contain complete Gaussian pairs. Neutral L marks still have no internal pair in their actual marked subinterval. Larger Wick exclusions and Gaussian-chaos projections retain their actual regions.

For a pairing π, let i_a be the number of internal chords in charged mark a, and zero for an L mark. Put $I(z) = \sum_a i_a z_a$, with commuting $z_a^2 = 0$. The old charge-tilt formula on compact complex source domains [2] suggests a conditional rule, but does not itself prove convergence at the single lattice charge m. Here we perform that projection explicitly, following the Fourier approach to fixed-charge complex SYK in [1] and retaining the marked-source cancellations.

*AI research assistant (OpenAI).

2 Exact singleton cancellation before Fourier projection

At one site write $n = c^\dagger c$, $\bar{n} = 1 - n$. The symmetric local body loss gives

$$(1 - zD)n = (1 - z)n, \quad (1 - zD)\bar{n} = \bar{n} + zn, \quad (1 - zD)c^{(\dagger)} = (1 - z/2)c^{(\dagger)}.$$

The second identity is crucial: an occupation with zero unmarked weight can acquire a pure source term. Thus an ordinary mark cannot be treated by deleting the no-internal-pair assertion from the pure Slater algorithm.

For a Fourier angle θ, use the normalized complex product trace with $\rho_\theta = e^{i\theta}/(1 + e^{i\theta})$. An internal charged chord has the two singleton ratios

$$(1 - z_a)(1 + e^{i\theta} z_a) = 1 + (e^{i\theta} - 1)z_a.$$

Each label appears p times. The complete ideal singleton residual is

$$\exp_{\mathrm{nil}}\{p(e^{i\theta} - 1)I(z)\}. \tag{1}$$

Open endpoints cancel the pd_a number source exactly; L singleton deficits cancel in the same way, without an internal-chord term. The scalar singleton normalization and the unmarked charge trace give

$$[\cos(\theta/2)]^{N-2kp}, \tag{2}$$

for k Gaussian chords. This factor must be kept until the extensive source terms have cancelled. It tends to a Gaussian after the change of variable $t = p\theta$.

Let $F_{\pi,N}(\theta, z)$ denote the normalized contribution of this pairing to the full-Fock trace with $e^{i\theta(Q-N/2)}$ and the indicated deficit sources. Include C_m^{-k} and the disorder normalization. Per chord, the combined label-count and source-free singleton normalization is

$$A_N = \frac{4^{-p}\binom{N}{p}^2}{\binom{m}{p}\binom{m+p}{p}} \longrightarrow e^{-\lambda}.$$

It is uniformly bounded; the remaining overlap factors are treated below. For every fixed real t, the normalized collision expansion gives, coefficientwise,

$$F_{\pi,N}(t/p, z) \longrightarrow e^{-t^2/(8\lambda)} \exp_{\mathrm{nil}}\{\lambda K_\pi(z) + itI(z)\}. \tag{3}$$

K_π is the existing full-trace source exponent, including $K_\pi(0) = -4\,\mathrm{cr}\,\pi$ and any Gaussian-number shifts. To obtain (3), divide each overlap-site polynomial by its source-bearing singletons. On compact t intervals their coefficients tend to their half-occupation values. Fixed label-pair overlaps have uniformly bounded exponential moments, and triple memberships vanish after domination. Formula (1) tends to e^{itI}. This extends the previous one/two-source argument to any fixed source count: the coefficient norm merely acquires a fixed polynomial power in the total overlap count.

3 An integrable bound for the full Fourier interval

We give the missing bound that permits conditioning on a single charge. Let R be the sum of pairwise intersection sizes of the $2k$ independent p-set labels. For fixed k and every fixed positive a,

$$\sup_N \mathbb{E}e^{aR} < \infty.$$

Indeed each pair overlap Y obeys $\mathbb{E}e^{aY} \le \exp[\lambda_N(e^a - 1)]$ by its factorial moments; Holder handles their finite sum without assuming independence.

Fix $0 < \delta < \pi/2$ and r source variables. On $|\theta| \le \delta$, the singleton denominators have nonzero constant terms uniformly away from zero. At most $2R$ label memberships are missing from ideal singleton sites. After the exact cancellation in (1), all corrections from those missing sites and overlap-site matrices have coefficient norm at most

$$K(1+R)^r A^R.$$

This follows by expanding the fixed-r nilpotent products: the local matrix maps have uniformly bounded coefficients, the number of non-singleton sites is at most R, and at most r marked factors may be chosen. No division by the possibly zero unmarked overlap numerator is used. The residual (1) adds at most $K(1+p|\theta|)^r$. Consequently each coefficient satisfies

$$|[z_S]F_{\pi,N}(\theta, z)| \le K(1+p|\theta|)^r |\cos(\theta/2)|^{N-2kp}. \tag{4}$$

The overlap average has been absorbed into K. Constants here depend on the fixed diagram, r and a compact positive range of λ_N. For sufficiently large N, $N - 2kp \ge N/2$, so under $t = p\theta$,

$$|[z_S]F_{\pi,N}(t/p, z)| \le K(1+|t|)^r e^{-t^2/(16\lambda_+)}. \tag{5}$$

For $\delta \le |\theta| \le \pi$, avoid complex singleton denominators. Every label configuration has support U at most $2kp$ sites. Loss on sites outside that support is zero because $D_i I = \mathcal{L}_i I = 0$. Sites outside the support therefore contribute exactly a factor of modulus $|\cos(\theta/2)|^{N-U}$. Direct expansion of the r loss and number factors gives at most Kp^r on the support. The normalized label count contributes at most e^{Kp}, since k is fixed. Thus

$$|[z_S]F_{\pi,N}(\theta, z)| \le Kp^r e^{Kp} |\cos(\theta/2)|^{N-2kp}. \tag{6}$$

Its integral, even multiplied by p, tends to zero exponentially in N. Equations (5) and (6) justify dominated convergence on the rescaled whole Fourier interval. This is stronger than compact-t convergence of a charge generating function.

4 The central source and the ordinary strong bridge

Exact half-charge extraction divides the integral of $F_{\pi,N}(\theta, z)$ by that of $\cos(\theta/2)^N$. Using (3)–(6), the limiting ratio is

$$e^{\lambda K_\pi(z)} \frac{\int_{\mathbb{R}} e^{-t^2/(8\lambda)} e^{itI(z)}\, dt}{\int_{\mathbb{R}} e^{-t^2/(8\lambda)}\, dt} = \exp_{\mathrm{nil}}\{\lambda[K_\pi(z) - 2I(z)^2]\}. \tag{7}$$

The Gaussian integral is a finite polynomial identity in the sources.

Theorem 1 (Fixed central ordinary source)**.** *For every fixed admissible marked layout, the half-filled sector limit is given by the full-trace source with*

$$K_{\pi,\mathrm{cen}} = K_\pi - 4 \sum_{a<b} i_a i_b z_a z_b.$$

All higher fixed central moments exist and are bounded along the double-scaling sequence. No source correction is added when a mark has no internal chords. Wick exclusions and chaos projections may be imposed before summing the finite set of diagrams.

For ordinary G sources, the explicit single-source coefficient is $-4\,\mathrm{cr}(O_a) - 2i_a$, where O_a is the open-chord set. The connected DD coefficient becomes $2b(b-1) - 4\,\mathrm{cr}(B)$, with $B = O_1 \cap O_2$, $b = |B|$. The former $4i_1 i_2$ term is absent. L has zero internal count, so the DL and LL formulas retain their previous values.

Normalize the actual ordinary field by the target-sector scalar:

$$G_{n,N} = C_m^{-n/2}[p\mathcal{N} - D - \lambda_N \mathcal{N}(\mathcal{N}-1)]H^n|_m.$$

Let $h^n = \sum_d a_{nd}(q) H_d^{(q)}(h)$, $q_N = e^{-4\lambda_N}$, $m_d(q) = q\partial_q \log[d]_q!$, and define

$$g_n^{\mathrm{cen}}(h) = \sum_d a_{nd}(q)[-4\lambda m_d(q) - \lambda(n-d)]H_d^{(q)}(h),$$
$$J_{d,N} = C_m^{-d/2}(dp - D)\mathcal{W}_d(H^d)|_m - \lambda_N[d(d-1) - 4m_d(q_N)]H_d^{(q_N)}(h_N).$$

In the finite-N g_n and a_{nd} below use q_N, λ_N. Cutting a fixed internal matching leaves its original q^{c_γ} weight. Its remaining source is $\lambda[-4\,\mathrm{inv}\,\sigma - 2i]$; its Mahonian average is $\lambda[-4m_d - 2i]$. Subtracting g_n^{cen} thus leaves the pure J source. The same grouping applies to RR, RJ and JJ Gram terms on the same original disorder, exactly as in [3], now with the central rule proved independently. Therefore, for fixed n,

$$\left\| G_{n,N} - g_n^{\mathrm{cen}}(h_N) - \sum_d a_{nd}(q_N) J_{d,N} \right\|_{2,m} \longrightarrow 0. \tag{8}$$

All fixed higher moments are bounded by the source theorem. Interpolation upgrades (8) to every fixed finite Schatten exponent; Holder permits fixed left/right energy powers. This is a same-copy strong statement, not equality of two marginal laws.

5 Actual pinching, selected preparations and a scalar calibration

The counterterm $g_n^{\mathrm{cen}}(h_N)$ commutes exactly with h_N. The preceding higher-Wick stage [4] established $\|\mathcal{P}_{h_N} J_{d,N}\|_{r,m} \to 0$ for every fixed d,r, using the central two-energy measure and its zero diagonal mass. Apply the contraction of exact pinching to (8). Then

$$\left\| (1 - \mathcal{P}_{h_N}) G_{n,N} - \sum_d a_{nd}(q_N) J_{d,N} \right\|_{r,m} \longrightarrow 0 \tag{9}$$

for every fixed n and finite r. Complete degenerate energy blocks are retained; no long-time averaging limit is interchanged.

For an existing fixed-degree stabilizer-covariant preparation $P_N O$ with bounded norm, the orbit-density estimate [5] gives

$$\mathbb{E}\,\|X_N P_N O\|^2 \le C_P\,\|X_N\|_{4,m}^2\,.$$

Thus (9) holds on such selected preparation vectors. Polarization, higher moments and trace Holder transfer any fixed covariant block between polynomial preparations. Fixed real h-time phases are handled by the already summable Gaussian energy Taylor tails. In particular, all fixed ordinary charged truncations now have an actual, exactly pinched selected-state limit. The limit uses the charged Wick tower's density $f(h) = 1 - \sqrt{q} + \sqrt{q}\,h^2$.

An exact finite-N check exposes the role of internal contractions. For m=N/2 the covariance ratio is

$$\frac{C_{m-1}}{C_m} = \frac{(m-p)(m+p+1)}{m(m+1)}, \qquad \frac{\mathbb{E}G_{2,N}^{\text{unnormalized}}}{C_m} = m\left(\frac{C_{m-1}}{C_m} - 1\right) = -\frac{p(p+1)}{m+1} =: b_{N,p}.$$

This tends to -2λ. At n=2 the full difference in (8) is exactly the deterministic scalar $(b_{N,p} + 2\lambda_N)I$, since $Z_2|_m / C_m = h_N^2 - 1$. For $N = 4p^2$,

$$b_{N,p} + 2\lambda_N = \frac{1 - 2p}{4p^2 + 2} \longrightarrow 0.$$

Two separately internal G2 blocks have a central diagram weight $4\lambda^2$. Their full-trace weight is $4\lambda^2 + 4\lambda$. The difference is the shared charge variance removed by (7); deleting it without a charge-conditioning argument would have been unjustified.

This closes the fixed-degree ordinary strong bridge. The full selected bilocal also needs a uniform finite-N bound on the tail as n grows. The constants in (4) depend on the fixed diagram and cannot be used as such a bound. The known full-Fock Clifford Riesz estimate [6] does not automatically apply to fixed-charge trace. Neither the Fourier argument nor preparation-density bounds authorize interchanging the infinite ordinary sum with the N limit.

6 Finite evidence and continuation

Three fresh outputs support separate parts of the construction. The exact algebra checker verifies 2,186 nilpotent Fourier transforms, 20 energy regressions, 96 RR/RJ/JJ Gram equalities across 32 layouts, two nonzero full-versus-central G2 gaps, and five finite-N G2 scalar bridges. It also saves 30 exact conditional-singleton coefficients at $N = 4p^2$, $p = 16, 32, 64, 128, 256$, including their finite-size errors.

The first checker had a coverage defect: its requested two-mark local loop generated no cases. That output is retained and supplies no local-source coverage. A separate checker explicitly enumerates the intervals and requires positive coverage. It verifies 3,496 two-source, 459 three-source and 1,265 four-source cases against the unrestricted-site matrix source engine, preserving 1,380 cases with internal pairs and 4,010 negative nonconstant coefficients.

For disjoint singleton supports, set $L = N - 2kp$ and take r internal source counts $i_1, \ldots, i_r$. The exact normalized half-charge coefficient is

$$\frac{p^r \prod_a i_a}{\binom{L}{L/2}} \sum_{j=0}^{r} (-1)^{r-j} \binom{r}{j} \binom{L}{L/2 - j}.$$

It tends to zero for odd r and to $(-4\lambda)^{r/2}(r-1)!! \prod_a i_a$ for even r. This isolates the conditioning factor; it is not the complete interacting diagram with overlap corrections. An independent Fourier quadrature checks ten finite systems against these exact rational values at 60 and 90 decimal digits. A Gaussian incomplete-integral bound controls the omitted $|t| > 16$ interval. Quadrature and roundoff are assessed by precision comparison, not interval certification.

The next task is the central finite-N ordinary degree tail. Sectorwise pure-Wick tagged bounds and covariance CP estimates remain available; the Riesz-dependent product step needs replacement or another uniform argument. The existing full-trace bilocal four-point result is not being redone. Spatial/mass generators and finite-G dynamics remain open. No new external publication is made in this stage.

References

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[2] Alan Day and Chengheng, *Joint charge tilt*, local proof, 7 November 2026; `susskind-memory-charge-tilt-proof.md`.

[3] Alan Day and Chengheng, *Joint memory regression*, local proof, 7 November 2026, Sections 3–5; `susskind-memory-joint-regression-proof.md`.

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[6] L. Ben Efraim and F. Lust-Piquard, *Poincare type inequalities on the discrete cube and in the CAR algebra*, arXiv:math/0702233v1, Section 6.4, Theorem 6.8.